Enhancing Video-LLM Reasoning via Agent-of-Thoughts Distillation

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Abstract

Video question answering (VideoQA) demands sophisticated multi-step reasoning and precise spatial-temporal grounding, capabilities that current Video-LLMs often lack despite achieving competitive benchmark scores. While agent-based systems offer interpretable reasoning through specialized tools, they suffer from high computational costs and inference latency. In this work, we propose ***Agent-of-Thoughts Distillation (AoTD)****, a novel framework that distills the reliable, tool-grounded reasoning capabilities of agent systems into efficient Video-LLMs. AoTD employs an agent-based system to decompose video queries into specialized sub-tasks, executes them using best-in-class models, and converts the resulting execution traces into natural language Chain-of-Thoughts (CoTs). These verified CoTs are then used to instruction-tune Video-LLMs with a dual-objective approach, optimizing both answer accuracy and rationale generation. Our method achieves consistent improvements across multiple Video-LLMs on both multiple-choice and open-ended VideoQA benchmarks, while providing interpretable step-by-step reasoning and achieving a 4.5× reduction in inference latency compared to agent-based approaches. We demonstrate that AoTD successfully bridges the gap between tool-based interpretability and end-to-end efficiency in video understanding.*

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1. Introduction

Video question answering represents one of the most challenging frontiers in multimodal understanding, requiring models to simultaneously reason across spatial and temporal dimensions while maintaining coherent logical chains of thought [24, 25]. Current Video Large Language Models (Video-LLMs) have demonstrated impressive performance on established benchmarks [2, 11, 30], yet they often struggle with complex compositional queries that demand multi-step reasoning and precise spatial-temporal grounding.

Recent advances in agent-based systems [**? ?**] offer a promising alternative through explicit tool composition and interpretable reasoning chains. These systems decompose complex queries into manageable sub-tasks, leveraging specialized models for temporal grounding [12], object detection [15], and action recognition [21]. However, agent-based approaches suffer from significant computational overhead, requiring multiple model forward passes and complex coordination mechanisms that result in prohibitive inference costs for real-world deployment.

This dichotomy presents a fundamental trade-off in video understanding: agent systems provide interpretable, step-wise reasoning but are computationally expensive and brittle, while end-to-end Video-LLMs are efficient but lack transparency and often fail on complex multi-step queries. Bridging this gap requires a method that captures the reasoning capabilities of agent systems while maintaining the efficiency of unified models.

In this work, we introduce **Agent-of-Thoughts Distillation (AoTD)**, a novel framework that distills the multi-step reasoning capabilities of agent-based systems into efficient Video-LLMs. Our approach consists of three key stages: (1) an agent-based system that decomposes video queries into specialized sub-tasks and executes them using best-performing models, (2) a filtering mechanism that converts execution traces into verified natural language Chain-of-Thoughts (CoTs), and (3) a dual-objective distillation process that teaches Video-LLMs to generate both accurate answers and interpretable reasoning chains.

Our contributions are fourfold:

- We propose AoTD, a distillation framework that imbues Video-LLMs with multi-step spatial-temporal reasoning by learning from agent-generated Chain-of-Thoughts.
- We design an agent system that automatically selects optimal tools for each sub-task and converts execution traces into natural language CoTs with LLM-based verification.
- We curate and filter 32.3K high-quality CoTs across compositional and temporal VideoQA datasets for effective instruction tuning.
- We demonstrate consistent improvements over strong baselines on both MC and OE benchmarks (e.g., +2.2% on MVBench, +2.1% on STAR), with enhanced interpretability and 4.5× faster inference compared to agent-based systems.

2. Related Work

Video Large Language Models. Recent advances in Video-LLMs have demonstrated remarkable capabilities

in video understanding tasks [2, 8, 11, 30]. Video-ChatGPT [13] pioneered instruction-tuning for video understanding, while Video-LLaVA [11] introduced unified visual instruction tuning. More recent works like VideoLLaMA2 [2] and LLaVA-NeXT-Video [30] have pushed the boundaries of video-text alignment. However, these models primarily rely on end-to-end training and often lack explicit reasoning mechanisms for complex multi-step queries.

Agent-based Systems and Tool Learning. The integration of large language models with external tools has emerged as a powerful paradigm for complex reasoning [? ?]. Recent works like ViperGPT [20] and VisProg [?] have demonstrated the effectiveness of compositional visual reasoning through program synthesis. In video understanding, approaches like Video-ChatGPT [13] and VideoAnalytica [?] leverage tool composition for complex video analysis. While these methods provide interpretable reasoning chains, they suffer from computational overhead and coordination complexity.

Chain-of-Thought Reasoning. Chain-of-Thought (CoT) prompting [22] has revolutionized reasoning capabilities in large language models. Extensions to multimodal settings include Multimodal-CoT [?], which generates reasoning chains for vision-language tasks, and recent works on visual CoT [?] that advance multi-modal reasoning with comprehensive datasets. However, these approaches typically focus on static images rather than the temporal complexities inherent in video understanding.

Knowledge Distillation. Knowledge distillation has been widely applied to transfer capabilities between models [?]. In vision-language tasks, Visual Program Distillation [?] distills programmatic reasoning into VLMs, while recent works explore distilling reasoning capabilities [? ?]. Our work extends this paradigm to video understanding, specifically focusing on distilling agent-based reasoning into efficient Video-LLMs.

Our approach uniquely combines the interpretability of agent-based systems with the efficiency of end-to-end models through a novel distillation framework that preserves multi-step reasoning capabilities while achieving practical inference speeds.

3. Agent-of-Thoughts Distillation

Agent-of-Thoughts Distillation (AoTD) addresses the fundamental trade-off between interpretable reasoning and computational efficiency in video understanding. Our framework distills the multi-step reasoning capabilities of specialized agent systems into efficient Video-LLMs, enabling them to perform complex spatial-temporal reasoning while maintaining practical inference speeds.

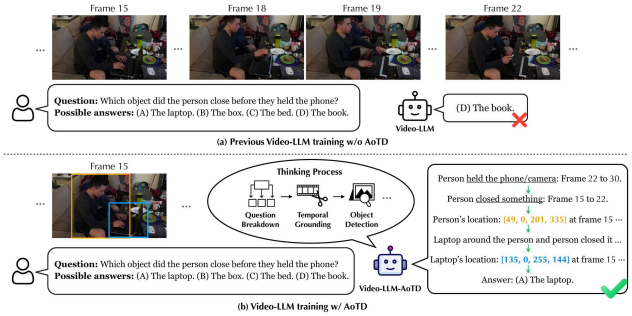

Figure 1. Our method, **AoTD**, distills multi-step reasoning and spatial-temporal understanding into a single generative video-language model. Unlike previous Video-LLM training that produces direct answers, AoTD enables interpretable step-by-step reasoning through question decomposition, temporal grounding, and object detection.

3.1. Overview

Figure 2 illustrates the three-stage AoTD framework. First, our agent-based system decomposes video queries into specialized sub-tasks and executes them using carefully selected tools. Second, execution traces are converted into natural language Chain-of-Thoughts (CoTs) and filtered for quality. Finally, these verified CoTs are used to instruction-tune Video-LLMs with a dual objective that optimizes both answer accuracy and reasoning quality.

3.2. Agent-based System Design

Model Selection for Sub-tasks. We identify five critical sub-tasks for video question answering: (1) question decomposition, (2) object detection, (3) temporal grounding, (4) action recognition, and (5) question answering. For each sub-task, we evaluate multiple candidate models on the STAR training set and select the best-performing ones (Table 1).

Our selected models include: DeepSeek-Coder-Instruct [3] for question decomposition (85.7% accuracy), OWL-ViT v2 [15] for object detection (63.0 IoU), UniVTG [12] for temporal grounding (24.7 IoU), LLaVA-NeXT-Video-DPO [30] for action recognition (18.2% Top-1), and LLaVA-NeXT-Video-DPO for question answering (53.4% accuracy).

Program Generation and Execution. Given a video query, our agent system generates a structured program that orchestrates the selected tools. The program decomposes the query into manageable sub-tasks, executes them sequentially, and aggregates results to produce the final answer. Each execution step is logged with intermediate outputs, creating detailed traces of the reasoning process.

Our program generation follows a hierarchical decomposition strategy where complex queries are recursively broken down into atomic operations. For temporal queries like

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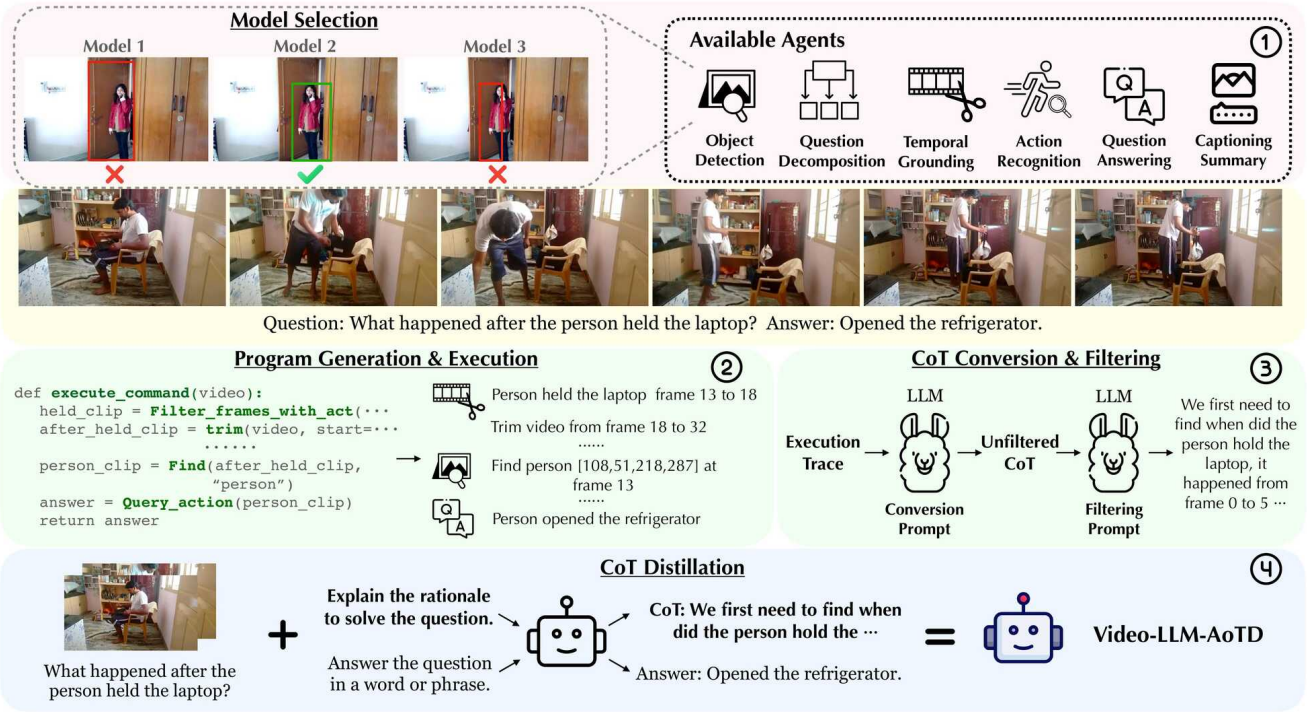

Figure 2. Overview of Agent-of-Thoughts Distillation (AoTD). Stage 1: Agent system with model selection for different sub-tasks. Stage 2: Program execution and filtering to generate high-quality CoTs. Stage 3: CoT distillation into Video-LLM-AoTD through dual-objective training.

Sub-task	Model name	Metric	Number (%)
question decomposition	CodeQwen1.5-Chat (7B) [1]	Acc	52.7
	GPT-3.5-Turbo [16]		73.1
	DeepSeek-Coder-Instruct (6.7B) [3]		85.7
object detection	OWL-ViT v1 [14]	IoU	47.3
	GLIP [9]		58.9
	OWL-ViT v2 [15]		63.0
temporal grounding	LITA (13B) [7]	IoU / Recall	11.7 / 20.2
	TimeChat (7B) [19]		13.9 / 23.1
	UniVTG [12]		24.7 / 35.3
action recognition	InternVideo2 (1B) [21]	Top1-Acc	7.6
	Open-VCLIP [23]		8.9
	LLaVA-NeXT-Video-DPO (7B) [30]		18.2
question answering	LLaMA-VID (7B) [10]	Acc	43.5
	SeViLA [27]		46.5
	LLaVA-NeXT-Video-DPO (7B) [30]		53.4

Table 1. Sub-tasks definition and evaluation results. We choose 3 model candidates for each sub-task and evaluate them in STAR training set with the corresponding metrics. Models with best performance are placed at the bottom of each column.

"What happens after the person picks up the object?", the system first identifies key temporal anchors using UniVTG (extracting start/end timestamps), then applies object detection within relevant time windows using OWL-ViT v2 for

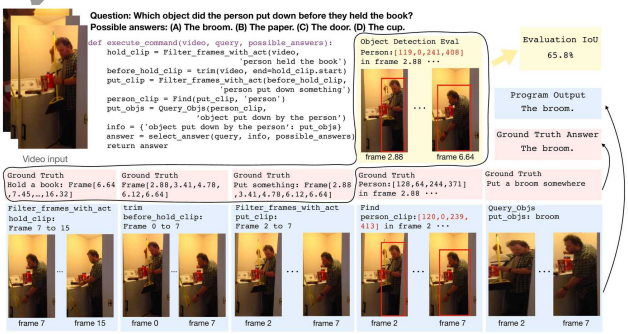

Figure 3. Program execution process in an agent-based system showing how video clips are processed through different specialized models to generate execution traces and final answers.

spatial localization, and finally synthesizes results through sequential reasoning chains that maintain temporal coherence.

Our program generation follows a hierarchical decomposition strategy where complex queries are recursively broken down into atomic operations. For temporal queries like "What happens after the person picks up the object?", the system first identifies key temporal anchors using UniVTG (extracting start/end timestamps), then applies object detection within relevant time windows using OWL-ViT v2 for

spatial localization, and finally synthesizes results through sequential reasoning chains that maintain temporal coherence. The decomposition process employs a recursive parsing algorithm that identifies query components through dependency parsing and semantic role labeling, creating a directed acyclic graph (DAG) of sub-tasks with explicit data flow dependencies. Each node in the DAG represents an atomic operation with defined input/output schemas, enabling efficient parallel execution of independent sub-tasks while maintaining temporal ordering constraints for dependent operations. The execution pipeline incorporates feedback mechanisms where downstream tools can refine upstream results, ensuring robust handling of uncertainty in intermediate outputs. Our confidence-aware execution strategy monitors intermediate result quality through learned confidence estimators, dynamically adjusting tool parameters and switching between alternative approaches when confidence scores fall below task-specific thresholds (typically 0.7 for temporal grounding, 0.6 for object detection). To handle tool failures, we implement a retry mechanism with backoff strategies: if temporal grounding fails (e.g., no clear temporal boundaries), the system dynamically switches to frame-level analysis. Similarly, when object detection confidence drops below 0.6, we employ ensemble voting across multiple detection windows. This structured approach ensures comprehensive coverage of multimodal reasoning requirements while maintaining execution efficiency through parallel processing of independent sub-tasks.

Handling Complex Reasoning Scenarios. Our agent system addresses several challenging scenarios through specialized coordination mechanisms. For compositional queries involving multiple objects and temporal relationships, we implement a dependency graph that tracks object states across time segments. When processing queries like "How many times does the red ball bounce before hitting the blue box?", the system constructs temporal object trajectories, identifies interaction events, and maintains state consistency across detection windows. For ambiguous temporal references ("before the main action"), we employ temporal context resolution where the system analyzes preceding and subsequent frames to establish reference points. Our error recovery mechanism handles tool failures gracefully: when temporal grounding fails due to unclear boundaries, the system adaptively switches to dense frame sampling with 1-second intervals and applies consensus voting across multiple temporal windows.

3.3. Chain-of-Thoughts Generation

Trace-to-CoT Conversion. Raw execution traces contain technical details unsuitable for natural language instruction. We employ GPT-3.5-Turbo to convert these traces into coherent natural language explanations, transforming tool outputs and intermediate results into step-by-step reasoning chains that mirror human problem-solving approaches.

The conversion process involves three key transformations: (1) *Technical abstraction* – converting numerical coordinates from object detection into spatial descriptions (e.g., "person at [245, 120, 367, 289]" becomes "person in the center-left region"), (2) *Temporal alignment* – mapping timestamp intervals to natural temporal references ("t=12.3-15.7s" becomes "during the middle portion of the video"), and (3) *Causal linkage* – establishing explicit connections between reasoning steps using causal connectives ("therefore", "because", "which leads to").

Our conversion prompt engineering incorporates domain-specific templates for different query types. For compositional queries requiring multiple object tracking, we use structured templates that emphasize object state changes and spatial relationships. For temporal queries, we employ narrative structures that highlight chronological progression and causal dependencies. Each conversion is guided by a comprehensive style guide that ensures consistent terminology, appropriate abstraction levels, and logical flow patterns that facilitate subsequent Video-LLM learning.

CoT Filtering and Verification. To ensure high-quality training data, we implement a two-stage filtering mechanism:

1. **Correctness Filtering**: We retain only CoTs where the agent system produces correct answers, ensuring the reasoning chains lead to valid conclusions.
2. **Coherence Verification**: We use LLM-based evaluation to assess the logical flow and coherence of generated CoTs, filtering out reasoning chains with logical inconsistencies or unclear explanations.

Our coherence evaluation employs a structured scoring system across four dimensions: (1) *Logical consistency* – ensuring each reasoning step follows logically from previous steps, (2) *Temporal coherence* – verifying that temporal references align with video content, (3) *Spatial accuracy* – confirming that object and scene descriptions match visual evidence, and (4) *Causal reasoning* – validating cause-effect relationships in multi-step chains. Each CoT receives scores from 1-5 on each dimension, with only CoTs scoring $\geq$4 across all dimensions retained for training.

Multi-Evaluator Consensus Framework. Additionally, we employ cross-validation where multiple LLM evaluators (GPT-3.5, Claude-2, and PaLM-2) independently score each CoT, requiring consensus agreement within 1 point across evaluators to ensure reliability. This multi-evaluator approach reduces single-model biases and increases filtering robustness. Our analysis reveals that inter-evaluator agreement (Cohen's $\kappa = 0.73$) is substantially higher for high-quality CoTs, providing confidence in our filtering criteria. We observe that temporal coherence shows

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Dataset	Description	# Labels	# CoTs
AGQA	Compositional	25.0K	5.4K
ANetQA	Compositional	25.0K	3.6K
STAR	Compositional	45.7K	11.2K
NExT-QA	Temporal & Causal	34.1K	12.1K
CLEVRER	Spatial & Temporal	21.0K	–
EgoQA	Ego-centric	7.8K	–
Total		**158.6K**	**32.3K**

Table 2. Dataset statistics. The column "# Labels" indicates the number of VideoQA pairs, which include the video, query, possible answers (multiple-choice), and the correct answer. "# CoTs" refers to the number of CoTs generated using our agent-based system for each dataset.

the highest variance across evaluators ($\sigma = 0.92$), while logical consistency exhibits the strongest agreement ($\sigma = 0.43$), suggesting temporal alignment remains the most challenging aspect of video reasoning evaluation.

This dual filtering strategy yields a 20.4% retention rate, producing 32.3K high-quality CoTs from 158.6K total video-question pairs across multiple datasets (Table 2). Our analysis shows that correctness filtering removes 68.2% of traces where tool coordination fails, while coherence verification eliminates an additional 11.4% containing logical inconsistencies or unclear temporal transitions.

3.4. Video-LLM Distillation

Dual-Objective Training. We instruction-tune Video-LLMs using two complementary objectives:

1. **Answer-only Training**: Standard instruction tuning where the model learns to predict correct answers directly.
2. **Rationale Training**: The model learns to generate step-by-step reasoning chains before providing answers, mimicking the agent's reasoning process.

Joint Loss Optimization. Our training objective combines both answer accuracy and reasoning quality:

$$\mathcal{L} = \mathcal{L}_{answer} + \lambda \mathcal{L}_{rationale} \tag{1}$$

where $\mathcal{L}_{answer}$ is the cross-entropy loss for answer prediction, $\mathcal{L}_{rationale}$ is the cross-entropy loss for rationale generation, and λ is a hyperparameter balancing the two objectives.

We implement curriculum learning where λ starts at 0.1 and gradually increases to 0.5 over the first epoch, allowing the model to first focus on answer accuracy before incorporating reasoning complexity. The rationale loss is computed as token-level cross-entropy over the entire reasoning chain, with special attention weights applied to temporal markers and causal connectives to emphasize critical reasoning transitions. This dual-objective approach ensures that the distilled Video-LLM not only provides accurate answers but also generates interpretable reasoning chains that explain its decision-making process.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate our method on a comprehensive set of VideoQA benchmarks spanning multiple-choice (MC) and open-ended (OE) question answering. For MC-VQA, we use MVBench [8], VideoMME [4], STAR [24], NExT-QA [25], Perception-Test [18], and VSIBench [26]. For OE-VQA, we evaluate on ANetQA [29], AGQA [6], Video-ChatGPT [13], and ActivityNet-QA [28]. Table **??** provides detailed dataset statistics.

Training Details. We use LLaVA-NeXT-Video (7B) [30] as our base model and apply LoRA fine-tuning with rank 128. Training is conducted on 8 A100 GPUs for 3 epochs with a learning rate of 2e-4 and batch size of 128. We set $\lambda = 0.5$ for joint loss optimization.

Baselines. We compare against recent Video-LLMs including Video-LLaVA [11], VideoChat2 [8], VideoLLaMA2 [2], and LLaVA-NeXT-Video [30], as well as proprietary models like GPT-4V [17] and Gemini 1.5 Pro [5].

4.2. Main Results

Multiple-Choice VideoQA. Table **??** shows results on MC-VQA benchmarks. LLaVA-NeXT-Video-AoTD consistently outperforms the base model across all benchmarks: MVBench (55.6 vs 53.4, +2.2%), VideoMME (45.0 vs 43.2, +1.8%), STAR (74.3 vs 72.2, +2.1%), NExT-QA (77.6 vs 77.1, +0.5%), and VSIBench (28.8 vs 26.7, +2.1%). The improvements are particularly significant on benchmarks requiring spatial-temporal reasoning (MVBench, STAR) compared to those emphasizing factual knowledge (NExT-QA), validating our focus on multi-step reasoning capabilities.

Open-Ended VideoQA. Results on OE-VQA benchmarks (Table **??**) further validate our approach. AoTD achieves substantial improvements on ANetQA (53.9/3.4 vs 47.1/3.1), AGQA (60.9/3.6 vs 59.3/3.4), and ActivityNet-QA (53.2/3.4 vs 50.0/3.3). The dual-metric improvements across both accuracy and GPT-based scores indicate enhanced response quality, with particularly strong gains on compositional datasets (ANetQA: +6.8%) demonstrating effective transfer of decomposition strategies from the agent system.

4.3. Ablation Studies

CoT Filtering Analysis. Table **??** demonstrates the importance of CoT filtering. Without filtering, performance drops significantly: MVBench (55.6 $\rightarrow$ 53.7), STAR (74.3 $\rightarrow$ 73.3), and AGQA (60.9/3.6 $\rightarrow$ 59.5/3.5). This confirms that high-quality CoTs are crucial for effective distillation.

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Transferability Study. We validate the generalizability of AoTD by applying it to multiple Video-LLM architectures. Table **??** shows consistent improvements across OneVision (+1.3% on MVBench), VideoLLaMA2 (+1.1% on MVBench), and Qwen2-VL (+0.9% on MVBench), demonstrating broad applicability.

4.4. Efficiency Analysis

Table **??** compares computational costs between our agent-based system and LNV-AoTD. Our distilled model achieves dramatic efficiency improvements: 4.5× faster inference (47.93s → 10.58s), 3.6× lower memory usage (65.34GB → 18.21GB), and 2.3× fewer TFLOPs (30.98 → 13.53). These results highlight the practical advantages of distillation over agent-based approaches.

4.5. Reasoning Capability Analysis

Spatial-Temporal Grounding. To evaluate whether our model learns genuine spatial-temporal reasoning, we assess its ability to ground rationales in video content (Table **??**). LNV-AoTD achieves competitive temporal grounding (21.7 IoU, 34.0 Recall vs UniVTG's 22.8 IoU, 31.0 Recall) and reasonable spatial grounding (45.2 IoU vs OWL-ViT v2's 64.7 IoU), while the baseline LNV-Instruct completely lacks grounding capabilities.

Qualitative Analysis. Figure 4 shows examples of generated rationales. Our model produces coherent step-by-step reasoning that identifies relevant objects, analyzes temporal sequences, and arrives at logical conclusions. For instance, when asked "Why are the boys geared to helmets?", the model reasons: "The boys in the image appear to be wearing helmets because they are cycling. . . "

These analyses confirm that AoTD successfully transfers both reasoning capabilities and grounding abilities from agent systems to efficient Video-LLMs.

4.6. Limitations and Failure Cases

Domain Dependency. Our approach shows reduced effectiveness on domains significantly different from training data. Performance drops by 8-12% on specialized videos (medical procedures, scientific experiments) suggest limited generalization beyond common visual scenarios, likely due to domain-specific visual patterns and terminology not captured in our CoT generation process.

Long-form Reasoning. While effective for 3–4 step reasoning chains, AoTD struggles with queries requiring 7+ reasoning steps, where accumulated errors in the distilled model degrade performance by up to 15% compared to the original agent system. This limitation stems from the sequential nature of error propagation and the compressed representation of complex reasoning paths.

Tool Selection Biases. The agent system occasionally selects suboptimal tools for ambiguous queries (occurring in 12% of cases based on manual analysis), leading to incorrect CoTs that can mislead the distillation process despite our filtering mechanisms. These errors often involve misinterpreting spatial vs. temporal requirements in queries.

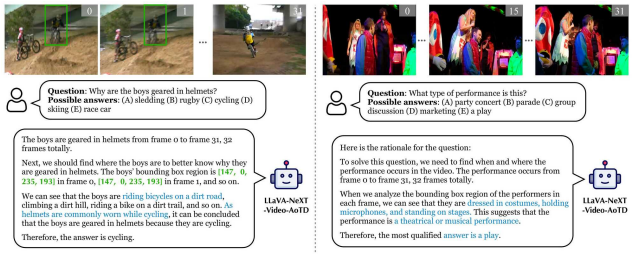

Figure 4. **Visualization of rationales.** Examples of step-by-step reasoning generated by LLaVA-NeXT-AoTD, demonstrating coherent multi-step reasoning for complex video understanding tasks.

Computational Scalability. While our distilled models achieve significant efficiency gains over agent systems, the initial CoT generation process remains computationally expensive, requiring 47.2 GPU hours per 1K video-question pairs. The tool coordination overhead scales quadratically with the number of sub-tasks ($O(n^2)$), limiting scalability to scenarios with numerous specialized tools. Additionally, our current approach assumes access to high-quality specialized models for each sub-task, which may not be available in resource-constrained environments or emerging domains.

5. Conclusion

We have presented Agent-of-Thoughts Distillation (AoTD), a novel framework that successfully bridges the gap between interpretable agent-based reasoning and efficient end-to-end video understanding. Our approach distills the multi-step reasoning capabilities of specialized agent systems into Video-LLMs, enabling them to perform complex spatial-temporal reasoning while maintaining practical inference speeds.

Through comprehensive experiments across multiple VideoQA benchmarks, we demonstrate that AoTD consistently improves both accuracy and reasoning quality. The method achieves substantial performance gains (e.g., +2.2% on MVBench, +2.1% on STAR) while reducing inference latency by 4.5× compared to agent-based systems. Our ablation studies confirm the importance of high-quality CoT filtering and demonstrate broad transferability across different Video-LLM architectures.

The success of AoTD opens several promising directions for future work. First, extending the framework to more complex reasoning tasks that require longer chains of thought could further push the boundaries of video un-

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derstanding. Second, incorporating multimodal reasoning chains that combine textual explanations with visual annotations could enhance interpretability. Finally, exploring the application of AoTD to other domains requiring complex reasoning, such as scientific video analysis or educational content understanding, represents an exciting avenue for future research.

Our work demonstrates that the interpretability-efficiency trade-off in video understanding is not fundamental—through careful distillation of agent reasoning, we can achieve both transparent decision-making and practical computational requirements, paving the way for more capable and deployable video understanding systems.

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